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YouTube Video Summarization & Q&A with RAG

AI-Powered, Context-Aware Summaries Using Transformer Models

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The Problem (Introduction):

- Over 500 hours of videos are uploaded to YouTube every minute.
- Viewers often struggle to extract key information from lengthy videos.
- Manual skimming is inefficient and time-consuming.
- There's a growing demand for:
 - Concise, high-quality summaries
 - Accurate answers to specific user queries

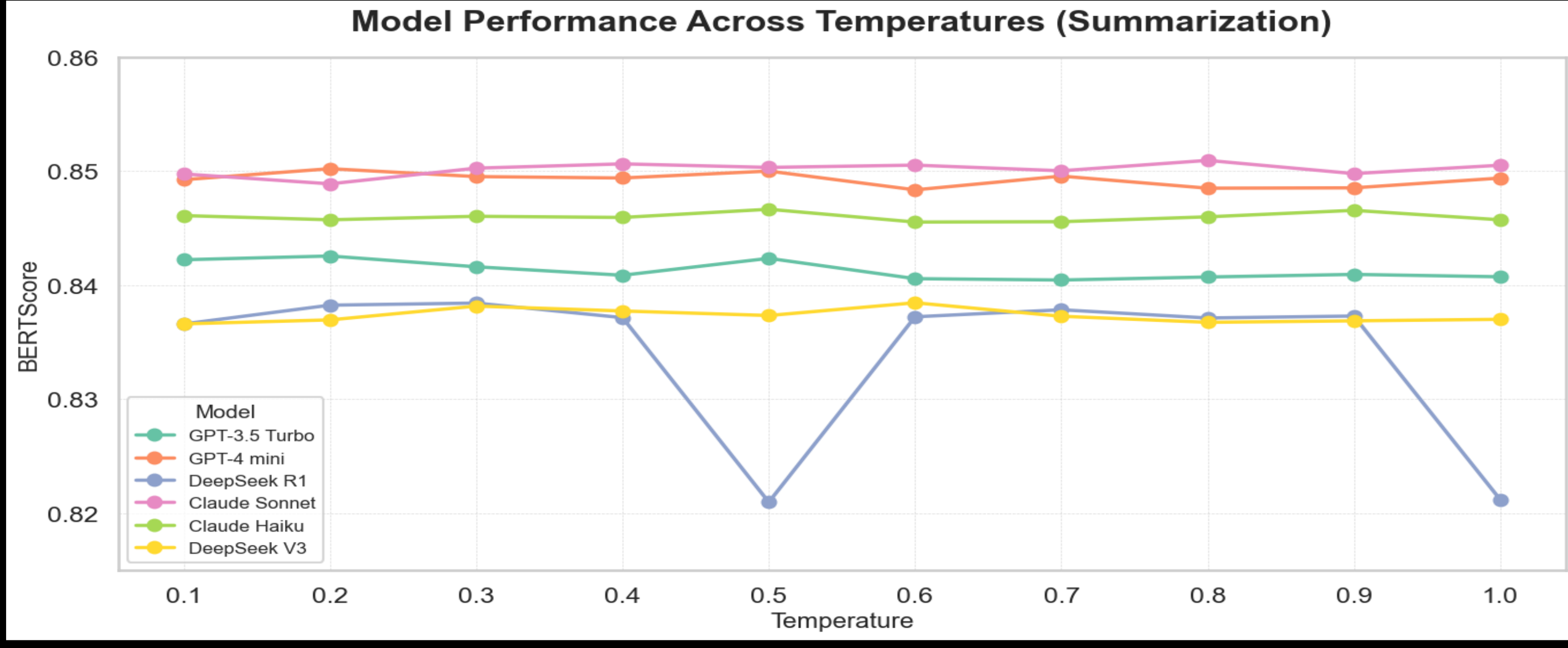
The Solution (Objective):

- Automatically generate summaries for YouTube videos.
- Provide precise answers to user-defined questions.
- Utilize Retrieval-Augmented Generation (RAG) to enhance contextual relevance.
- Leverage:
 - Transformer-based language models
 - Vector embeddings for efficient information retrieval

How it works (Method Overview):

- Transcript Extraction: Fetching video transcripts using the YouTube API.
- Chunking + Embeddings: Splitting transcripts into manageable chunks and converting the chunks into vector representations using embeddings.
- ChromaDB: Storing vectors in a local vector database for effectively retrieving the most relevant chunks for a query.
- Transformer Models: Using large language models to summarize the video content and generate answers to user questions.

Model Performance Across Temperatures (Summarization)

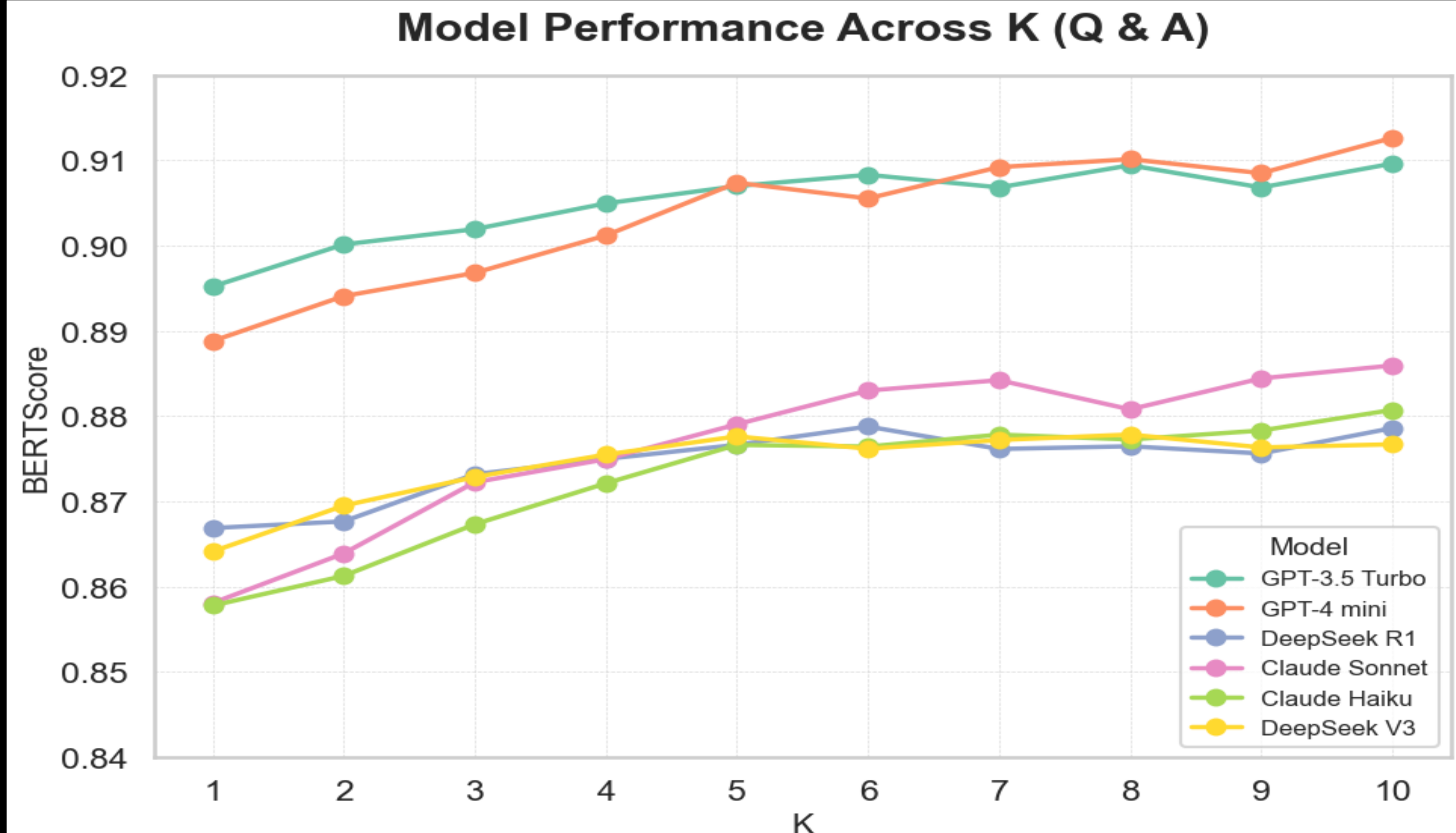


Model	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
GPT-3.5 Turbo	0.842	0.843	0.842	0.841	0.843	0.841	0.841	0.841	0.841	0.841
GPT-4 mini	0.849	0.850	0.850	0.849	0.850	0.848	0.850	0.848	0.849	0.849
DeepSeek R1	0.837	0.838	0.838	0.837	0.821	0.837	0.837	0.837	0.837	0.821
Claude Sonnet	0.849	0.848	0.850	0.851	0.850	0.851	0.850	0.851	0.850	0.851
Claude Haiku	0.846	0.846	0.846	0.846	0.847	0.846	0.846	0.846	0.847	0.846
DeepSeek V3	0.837	0.837	0.838	0.837	0.837	0.838	0.837	0.837	0.837	0.837

Experiment settings:

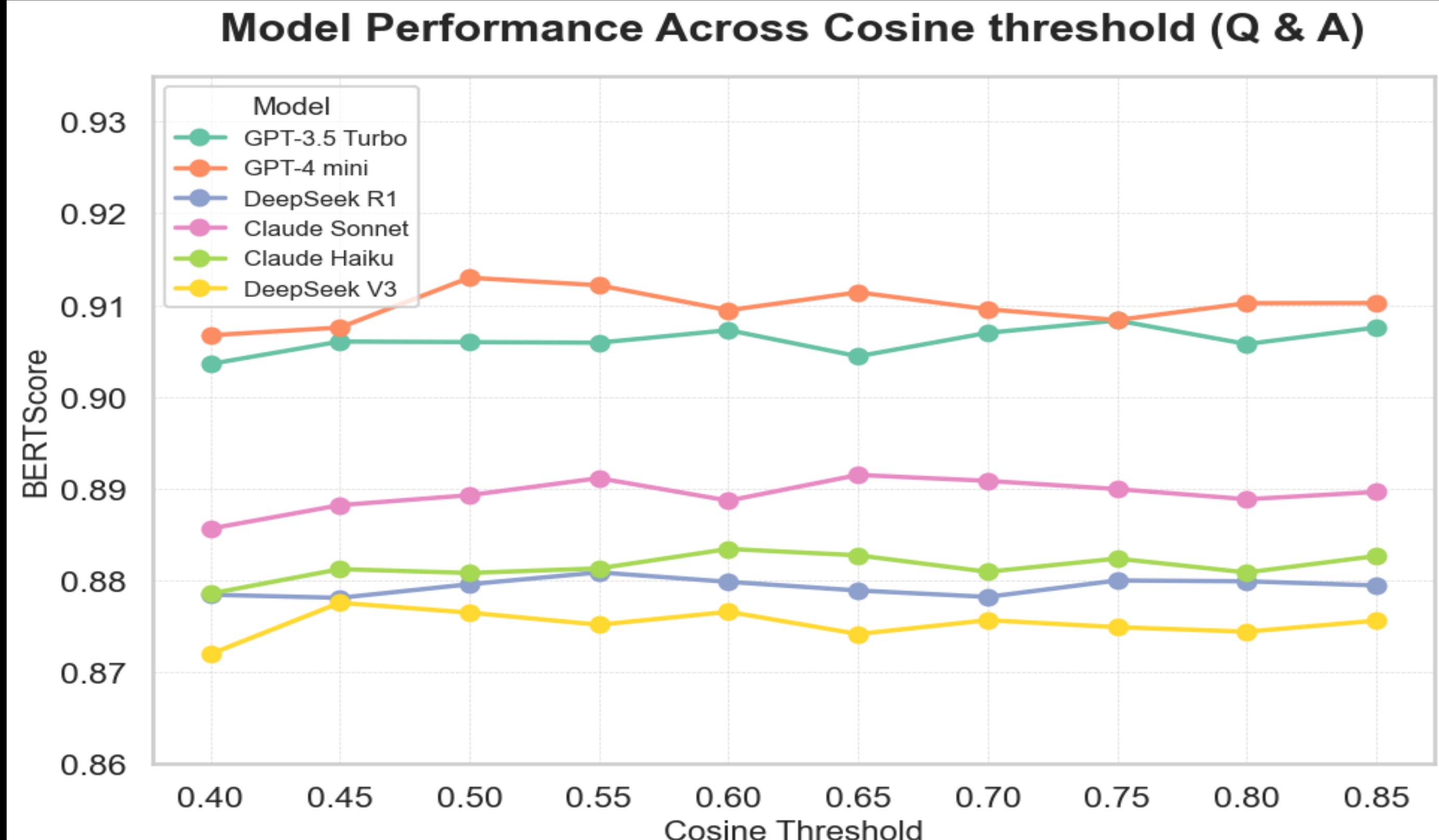
- Summarization: Models are evaluated at different temperature values for response variability and coherence.
- Q & A: tested using top K (number of chunks retrieved) and Cosine threshold (filters relevant content based on similarity score).

Model Performance Across K (Q & A)



Model	1	2	3	4	5	6	7	8	9	10
GPT-3.5 Turbo	0.895	0.900	0.902	0.905	0.907	0.908	0.907	0.910	0.907	0.910
GPT-4 mini	0.889	0.894	0.897	0.901	0.907	0.905	0.910	0.911	0.909	0.913
DeepSeek R1	0.867	0.868	0.872	0.875	0.877	0.879	0.877	0.877	0.876	0.879
Claude Sonnet	0.858	0.862	0.872	0.875	0.879	0.883	0.884	0.881	0.884	0.886
Claude Haiku	0.858	0.862	0.868	0.872	0.877	0.878	0.878	0.878	0.879	0.881
DeepSeek V3	0.864	0.869	0.873	0.875	0.877	0.876	0.877	0.877	0.876	0.877

Model Performance Across Cosine threshold (Q & A)



Model	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85
GPT-3.5 Turbo	0.904	0.906	0.906	0.906	0.907	0.905	0.907	0.908	0.906	0.908
GPT-4 mini	0.907	0.908	0.913	0.912	0.910	0.912	0.910	0.908	0.911	0.911
DeepSeek R1	0.878	0.878	0.880	0.881	0.880	0.879	0.878	0.880	0.880	0.879
Claude Sonnet	0.885	0.888	0.889	0.891	0.889	0.892	0.891	0.890	0.889	0.890
Claude Haiku	0.878	0.881	0.881	0.881	0.883	0.883	0.881	0.883	0.881	0.883
DeepSeek V3	0.872	0.878	0.877	0.876	0.877	0.874	0.876	0.875	0.875	0.876

Key takeaways:

- GPT-4 mini & GPT-3.5 Turbo lead in Q&A, with BERT scores > 0.90 at higher K values.
- Claude Sonnet shows stable performance across temperatures, maintaining ~0.85 BERT scores.
- As K increases the performance also increases consistently for all models and there is no particular trend for cosine threshold.

Models Evaluated:

Model	Type	Temper atue	K	Cosine Threshold	BERT Score (Summarization)	BERT Score (K) (Q & A)	BERT Score (Cosine) (Q & A)
GPT-3.5 Turbo	Chat Model	0.3	10	0.75	0.84255	0.90962	0.90832
GPT-4 mini	Chat Model	0.2	10	0.5	0.85020	0.91264	0.91298
DeepSeek R1	Open Source	0.3	6	0.55	0.83842	0.87879	0.88085
Claude Sonnet	Chat Model	0.8	10	0.65	0.85093	0.88595	0.89147
Claude Haiku	Chat Model	0.5	10	0.6	0.84664	0.88073	0.88341
DeepSeek V3	Open Source	0.6	8	0.45	0.83845	0.87785	0.87754

Dataset: ClarityClips/youtube-video-summarization (Hugging Face).

Q&A Benchmark: Manually curated synthetic dataset.

Metric: BERTScore for evaluating relevance and quality.

BERT Score Inference

1 – 0.95	Excellent
0.94 – 0.85	Very Good
0.84 – 0.7	Good
0.69 – 0.5	Fair

Conclusion:

This experiment demonstrates the effectiveness of combining Retrieval-Augmented Generation (RAG) with transformer-based models for summarizing and querying YouTube videos. Fine-tuning retrieval parameters like topK and cosine similarity significantly enhances Q&A performance. Stable summarization is achievable through careful temperature control, while retrieval thresholds influence both efficiency and accuracy.

Reference:

- Retrieval-Augmented Generation for Knowledge-Intensive NLP TasksPatrick Lewis, Ethan Perez, Aleksandra Piktus, et al.arXiv preprint arXiv:2005.11401 (2020)
- BERTScore: Evaluating Text Generation with BERTTianyi Zhang, Varsha Kishore, Felix Wu, et al.arXiv preprint arXiv:1904.09675 (2019).
- Document Q&A with ChromaDB - Google Generative AI Documentation.